

Contribution Details

L-RAPID: Lightweight Reversible And Privacy-preserving IIoT Data-perturbation system

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Long Abstract

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Abstract

IIoT(Industrial Internet of Things) systems constantly stream telemetry data from sensors and machinery to the cloud, posing serious privacy and security challenges. Conventional encryption methods are computationally too expensive for legacy devices, and noise-based perturbation methods irreversibly distort data. We introduce and evaluate a lightweight, fully reversible data perturbation pipeline for IIoT telemetry. To assess the balance between privacy and utility, a dataset comprising 31,106 IIoT intrusion detection samples covering five distinct attack categories was employed. After training a classifier that achieved a 98.47% binary classification accuracy, we evaluated its robustness and the resulting masking effects using four different perturbation techniques: Gaussian noise, Laplacian noise, XOR 16-bit, and XOR 64-bit.

IIoT equipment produces valuable operational telemetry data, the capture of which risks revealing commercially valuable information about manufacturing operations and system design. Complete cryptographic shielding is too expensive for legacy equipment, while the noise addition approach may introduce permanently distorted signals. There is a knowledge gap for a privacy technique that is cheap, robust to inference attacks, and entirely reversible so that authorized users can back-extract the true values. This work bridges the gap by evaluating four approaches on a realistic IIoT dataset with 3,000 normal samples and designing a rigorous composite scoring system for evaluation.

The dataset contains a mixture of 31,106 Modbus telemetry samples, correlated with other operational data to label them as under attack or benign: 15,000 normal (48%) and 16,106 attack samples (52%) across five attack categories: Injection, Backdoor, scanning, XSS, and Password. The feature register FC1FC is stationary, with variance ratios ranging from 0.56 to 0.58, indicating that the raw signal is information-rich and difficult to model externally.

The Random Forest Classifier achieved 98.47% (weighted F1: 0.9847) on a binary detection of normal versus attack, with 16 (of 3000) normal samples classified as attacks and 79 (of 3222) attack samples missed on the binary task. This provides evidence that the raw IIoT telemetry network has a highly exploitable discriminative structure in its features and simultaneously justifies privacy-preserving perturbation before transmission.

The study evaluated four distinct perturbation strategies. One such method, XOR masking, uses a bitwise exclusive-OR (XOR) operation to combine data values with an undisclosed secret key. Because this deterministic process preserves the data without introducing randomness, it remains entirely reversible. The 16-bit approach uses a 65,536 key space, while the 64-bit approach stretches it to 264. The testing phase also included additive noise utilizing Gaussian and Laplacian distributions. To guarantee consistency across these randomized methods, a uniform random seed was employed. The Laplacian distribution's "fatter" tails caused perturbed data points to be displaced significantly farther from their original positions than in the Gaussian distribution at equivalent power levels, thereby increasing the difficulty of statistical unperturbation for an adversary. While all four algorithms successfully achieved 100% exact-match restoration, they simultaneously hindered attackers' efforts, typically lowering accuracy to 47-51%. 50% adversarial accuracy in binary classification implies the attacker has no knowledge from which to infer the results, and thus the probability of each outcome is 0.5.

XOR 64-bit is the most advantageous perturbation technique among the methods evaluated. Its computational efficiency is high, requiring only a single CPU instruction per value, and it allows for deterministic data reconstruction without the need to store a seed. Furthermore, the massive keyspace makes brute-force recovery attempts impractical. All methods are shown to be totally invertible, resulting in no data loss irrespective of the type of perturbation applied, further demonstrating that total reversibility and a high degree of privacy are compatible. The uniformity of low attack accuracy seen by each method suggests that what is necessary to achieve secure communication is not a particular method of perturbation, but, in general, to eradicate the statistical structure of the transmitted signal, in whatever form, or by whichever of the introduced methods. Gaussian noise masking should be used only for initial feasibility analysis, as its additive nature and inherent predictability make it highly sensitive to statistical analysis. The 16-bit XOR is suitable for environments with highly constrained computational resources rather than for environments lacking key management.

This paper shows that the proposed lightweight, reversible perturbation can secure IIoT telemetry without compromising data utility for authorized receivers. All four tested methods achieved 100% reversibility and close to 50% adversarial accuracy. XOR 64-bit masking proved to be the optimal solution across privacy, security, and implementation simplicity. Further research should investigate adaptive key rotation to achieve forward secrecy, streaming perturbation for telemetry pipeline processing, and performance benchmarking under real-world conditions on physical IIoT boards.